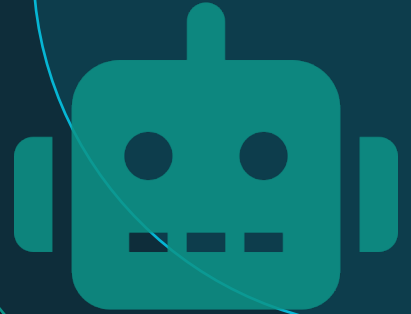


THE FUTURE IS ALREADY HERE

What if your AI could think, plan, and act... *without asking you first?*

Welcome to the era of Agentic AI Frameworks



The Office Analogy

 Imagine a company with 500 employees. Now imagine replacing the entire middle-management layer with AI agents that never sleep, never miscommunicate, and always optimize for the company's goals.



Traditional AI

Answers your questions.
Waits for instructions.
One task at a time.



Agentic AI

Sets its own sub-goals.
Uses tools independently.
Coordinates with other agents.



Multi-Agent AI

Teams of agents collaborate.
Specialist + manager roles.
Solves enterprise-scale problems.

What You'll Walk Away With

By the end of this session:



Understand WHY agentic AI is the next competitive frontier for every industry



Decode the architecture of LLM agents — Perception → Cognition → Action



Compare LangGraph, AutoGen & CrewAI — and know exactly when to use each



Navigate real challenges: bias, scalability, security, and ethical deployment



See the 2026 frontier — multi-agent systems, AI copilots, autonomous workflows

What Is Agentic AI?

AI systems that perceive the environment, reason about it, and take autonomous action – without constant human instruction.

 *Analogy: Like a self-driving car – it sees the road, plans the route, avoids obstacles, and arrives at the destination. You just say "Take me to work."*



Perceive

Reads text, images, sensors, APIs – makes sense of raw data



Reason

Plans, prioritizes, and decides what to do next



Act

Executes tasks, calls tools, collaborates with other agents



Perceive



Reason



Act

The 4 Superpowers of Agentic AI

Adaptability

Adjusts on the fly

Like a seasoned pilot — when weather changes, it reroutes. Reacts to new data, new constraints, new environments without being reprogrammed.

Decision-Making

Learns, improves, refines

Like a chess grandmaster who studies past games — it learns from outcomes and gets smarter with every interaction.

Autonomy

Works independently

Like a remote worker in a different timezone — keeps the workflow moving 24/7 without waiting for a manager to unblock tasks.

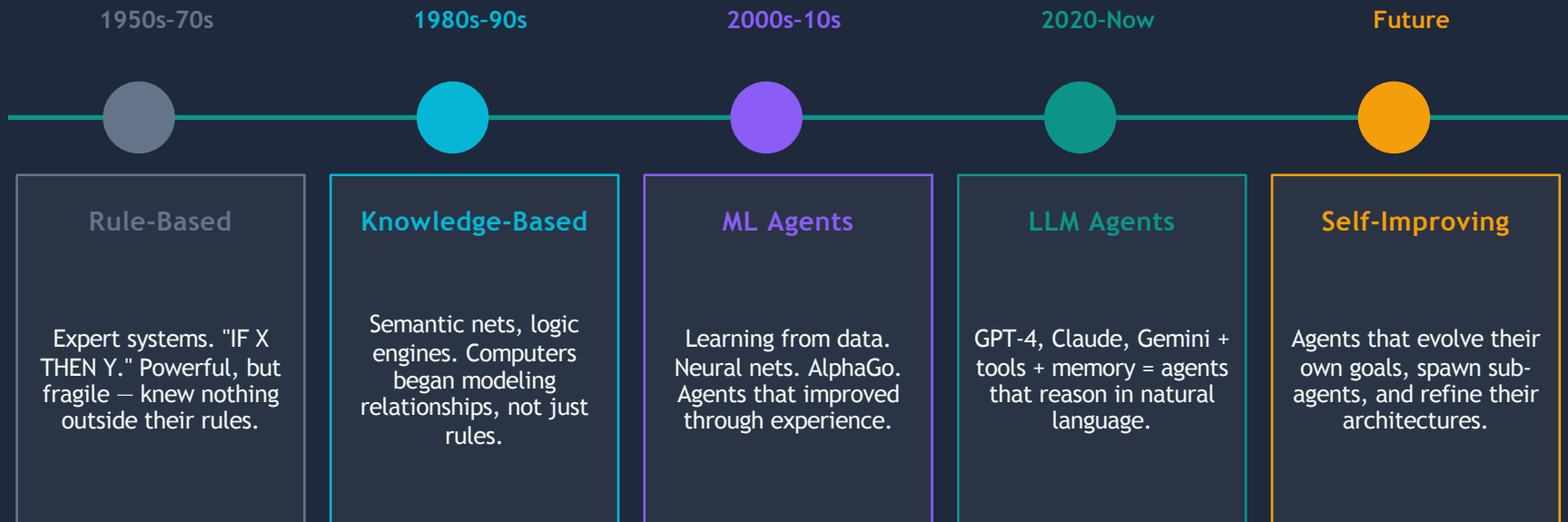
Collaboration

Agents working as teams

Like a hospital ER team — triage agent, diagnostic agent, treatment planner, all coordinating seamlessly to handle a complex case.

70 Years in 70 Seconds – The Evolution of AI Agents

From "if-then" rules to systems that rewrite themselves:



Traditional AI vs. Agentic AI

🔧 *Old way: A calculator that answers when asked.* 🚀 *New way: A strategist that acts without being asked.*

Feature	Traditional AI	Agentic AI
Control	Requires human oversight	✓ Fully autonomous
Decision-Making	Rule-based, predefined	✓ Dynamic, adaptive
Task Execution	Handles one task at a time	✓ Multi-tasking, collaborative
Learning	Static models	✓ Continuous, real-time learning
Scalability	Limited to specific domains	✓ Works across diverse domains

Where Agentic AI Lives — Today



AI Assistants

Perplexity, Claude, Copilot

Answer questions AND take actions — book meetings, write code, file bugs



Autonomous Vehicles

Waymo, Tesla FSD

Real-time sensor fusion + decision-making at 100mph — no human in the loop



Finance

Jane Street, Citadel AI

HFT bots making 10,000 decisions/second. Fraud detection that adapts in real time



Healthcare

Google DeepMind, Nabla

Diagnosing rare diseases from genomic data. AI agents that monitor patients 24/7



E-Commerce

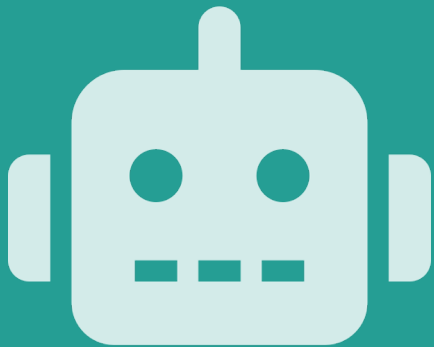
Amazon, Shopify AI

Agents that dynamically reprice, restock, and personalize for 300M customers simultaneously

SECTION 02

Inside the LLM Agent

How do they actually think, remember, and act?



Anatomy of an LLM Agent

Think of it as a brain with arms, memory, and a task list.



The Agent's Three Operating Modes

01

Perception Module

The Agent's Senses

- › Reads raw input: text, images, APIs, sensor streams
- › Filters noise — decides what matters vs. what to ignore
- › Converts unstructured data into structured signals
- › Real example: GitHub Copilot reading 10,000 lines of code context

02

Cognitive Engine

The Agent's Brain

- › Interprets signals and applies logical reasoning
- › Evaluates scenarios: "If X, then Y or Z?"
- › Learns from feedback loops — gets smarter over time
- › Real example: OpenAI's o3 reasoning chain on multi-step math

03

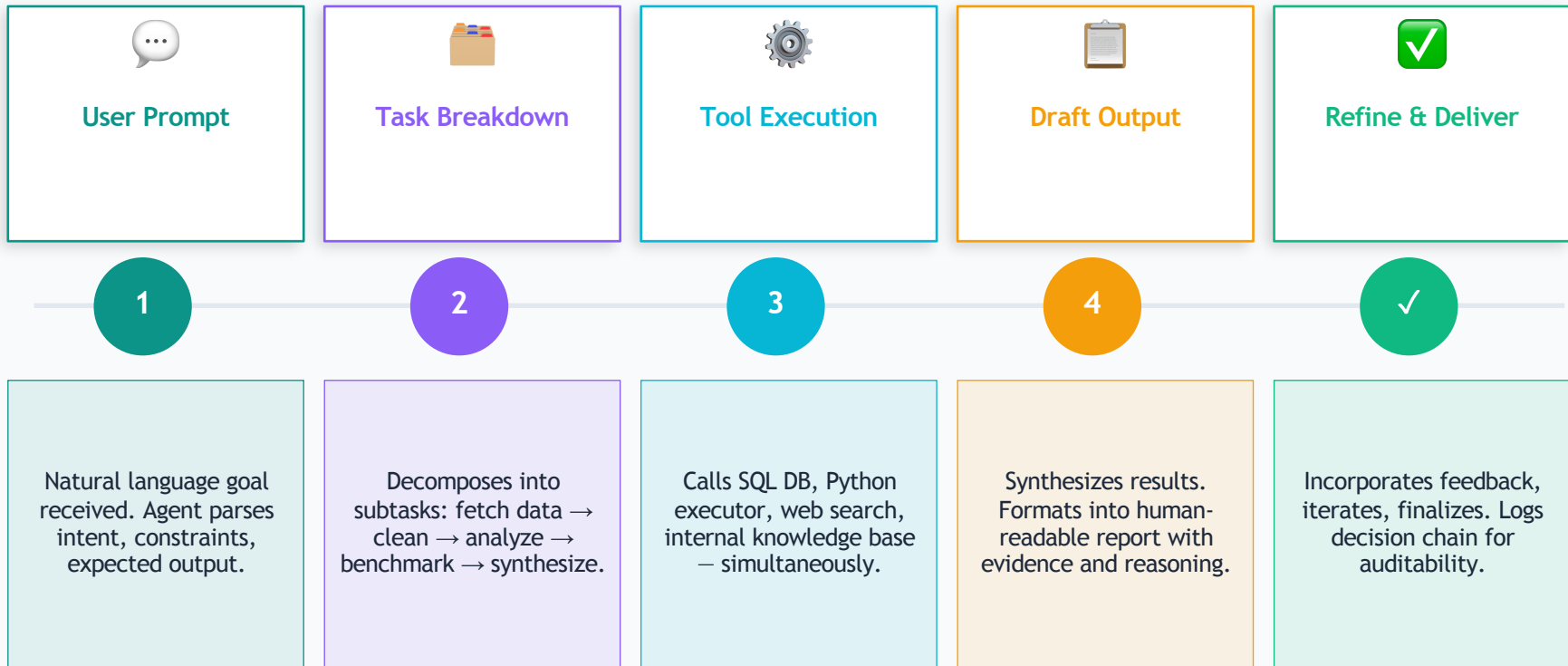
Action Module

The Agent's Hands

- › Executes decisions via tools, APIs, or direct output
- › Monitors outcomes and loops back for correction
- › Handles error states — retries, escalates, or stops
- › Real example: AutoGPT browsing the web, writing files, calling APIs

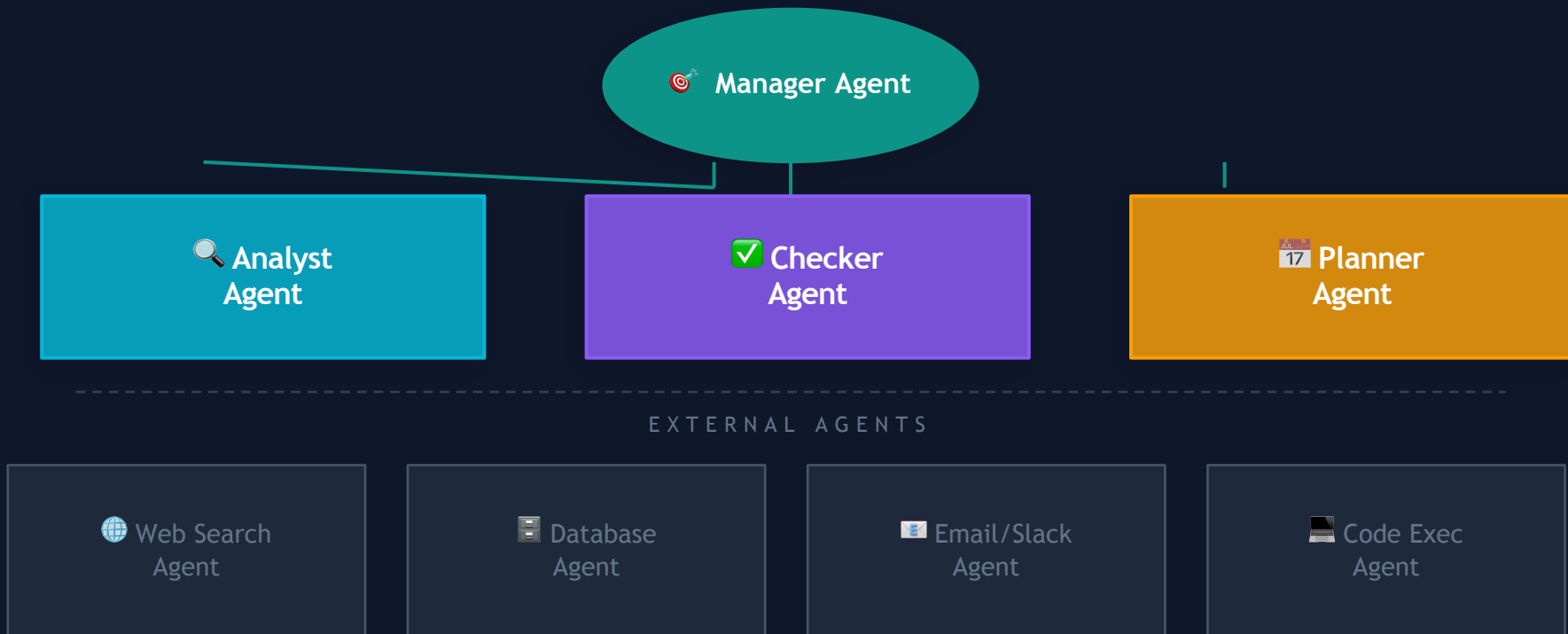
How an LLM Agent Handles Your Request

Scenario: "Analyze our Q1 sales data and suggest 3 strategic pivots for Q2"



Multi-Agent Teams – The New Org Chart

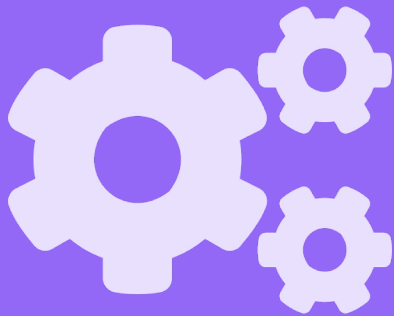
Modern AI systems work like companies: hierarchies, specialists, handoffs, and oversight.



SECTION 03

Pick Your Weapon

LangGraph vs. AutoGen vs. CrewAI — which one wins for your use case?



The Big Three – LLM Agent Frameworks



LangGraph

The Workflow Architect

Strength

Complex, stateful, looping workflows

Analogy

Like a GPS – tracks every turn, handles detours, maintains state

Best For

Finance, healthcare, logistics – anything with conditional logic and memory

Graph-based. Nodes = agents. Edges = conditions. State = shared memory.



AutoGen

The Debate Club

Strength

Multi-agent conversation and collaboration

Analogy

Like a boardroom meeting – multiple experts debate until the best answer emerges

Best For

Code review, research, content strategy, complex problem-solving with multiple roles

Conversation-driven. Agents chat with each other to refine solutions iteratively.



CrewAI

The Dream Team

Strength

Role-based, specialized agent teams

Analogy

Like hiring a consulting firm – each agent is a specialist with a defined role and deliverable

Best For

Market research, content pipelines, incident response – sequential/parallel specialist work

Role → Task → Tool model. Easiest to get started. Most intuitive for business users.

Framework Decision Matrix

"Which framework should I use?" — Answered.

	LangGraph	AutoGen	CrewAI
Scaling	Kubernetes operators	Azure autoscale	Celery + Redis
Optimization	Dask parallelization	Multi-LLM hot-swapping	Role-specific fine-tuning
State Mgmt	Versioned graph snapshots	Ephemeral sessions	Redis checkpoints
Security	JWT + API key vaults	Azure AD + RBAC	HIPAA/GDPR modules
Observability	OpenTelemetry dashboards	Azure Monitor + App Insights	Prometheus + ELK/Splunk
Complexity	High (graph expertise req'd)	Medium (conversation-first)	Low (role→task model)

Choosing the Right Framework — Real Scenarios



LangGraph

Loan Application Processing

Bank receives 1,000 loan apps/day. Agent verifies documents → checks credit scores → loops back if docs are missing → escalates to human reviewer if fraud is detected. State is preserved across the entire multi-hour process.



AutoGen

Automated Bug-Fixing Debate

Dev agent proposes a fix → Tester agent runs edge cases → Architect agent reviews design implications → Security agent checks for vulnerabilities. They debate until consensus. Final code is audited by 4 perspectives before merging.



CrewAI

Market Intelligence Report

Researcher agent scrapes 500 competitor sources → Analyst agent identifies 7 trend signals → Writer agent produces an executive brief → Editor agent checks for accuracy and bias. Delivered in 8 minutes. Used to take 2 analysts a full week.

The Hard Truths — Challenges You Must Know



Ethical Concerns

Biases in training data get amplified at scale. In 2025, an insurance AI agent denied 30% more claims from minority zip codes — the bias was invisible until it caused a lawsuit.

Fix: Bias auditing at every decision node. Human-in-the-loop for high-stakes decisions.



Scalability Gaps

A single LLM call costs \$0.002. A 10-agent workflow with 50 steps costs \$1+. At enterprise scale with 10,000 workflows/day — that's \$10K/day. Token efficiency is now a competitive advantage.

Fix: Smart caching, prompt compression, tiered model selection (GPT-4o mini for simple steps).



Performance Bottlenecks

Real-time agentic systems face latency challenges. A trading agent that takes 800ms to decide is worthless in HFT. Even customer-facing agents must respond in <2 seconds to avoid drop-off.

Fix: Streaming outputs, parallel agent execution, speculative decoding, edge deployment.



Security & Trust

Prompt injection attacks can hijack autonomous agents. In 2025, a red team fed a customer service agent a malicious user message that caused it to exfiltrate internal pricing data.

Fix: Input sanitization, output guardrails, sandboxed tool execution, audit trails.

Let's Make This Real – Demos & Discussion



3 Audience Questions

1. If you could delegate ONE complete workflow to an AI agent team today – what would it be and why?
2. What's the #1 thing that would need to be true for your organization to trust an autonomous AI agent with a customer-facing decision?
3. Between LangGraph, AutoGen, and CrewAI – which feels most natural for your team's skill set?



2 Live Demo Ideas

Demo 1 – CrewAI in 5 Minutes

- Define 3 agents: Researcher, Analyst, Writer
- Task: "Summarize today's top 3 AI news stories"
- Run live – watch agents hand off to each other
- Show the token usage & latency in LangSmith

Demo 2 – LangGraph Stateful Flow

- Build a 4-node approval graph: Submit → Verify → Approve/Reject → Notify
- Inject a 'missing document' to show the loop-back logic
- Visualize the graph in LangGraph Studio



Mini Case Study: Klarna (2025) – Deployed a 2-agent customer support system that handled 700 agents' worth of tickets in 1 month, saved \$40M annually, and improved customer satisfaction by 22%. Framework: custom LangGraph + OpenAI GPT-4o.

What's Next — The Frontier of Agentic AI



Adaptive Agent Networks

Self-optimizing agent swarms that reallocate compute, budget, and attention dynamically. Like a company that restructures itself every hour based on live KPIs.



Domain-Specialized Ecosystems

Healthcare agents integrating genomics + patient history + real-time diagnostics. Legal agents that reason about jurisdiction, precedent, and regulatory change simultaneously.



Human-AI Symbiosis

Agents that capture institutional knowledge — not just data, but context, judgment, and tacit expertise. Your most experienced employee's decision-making preserved forever.



Agentic Internet

Web agents that don't just browse — they negotiate, transact, and collaborate with other AI agents autonomously. The first phase of agent-to-agent economy is already underway.

What You Now Know

1

Agentic AI is not just better chatbots — it's autonomous systems that perceive, plan, and act to achieve goals, end-to-end.

2

The Perception → Cognition → Action loop is the engine powering every LLM agent — from simple chatbots to multi-agent enterprise systems.

3

LangGraph = complex state, AutoGen = collaborative debate, CrewAI = specialized roles. Right tool → 10x better outcomes.

4

Bias, scalability, latency, and security are real challenges — but solvable with architecture, governance, and human oversight.

5

The future belongs to builders who understand both the technology AND the responsible deployment of autonomous AI systems.

The industrial revolution
gave machines our muscles.

Agentic AI is giving machines
our judgment.

What we do with that — is still entirely up to us.

Thank you. Build something remarkable.